



University of Colombo School of Computing
SCS2312 Computational Models and
Programming Language Concepts
Tutorial - 06

- 1) Design a DFA that accept the all string start with '0' over $\{0,1\}$
- 2) Design a DFA that accept set of all string over $\{0,1\}$ of length 2
- 3) Design a DFA that accept any string over $\{a,b\}$ that does not contain the string 'aabb' in it
- 4) Construct a regular expression for the language of all strings over $\{a, b\}$ that contain at least one a .
- 5) Construct a regular expression for all strings over $\{0, 1\}$ that end with 01 .
- 6) Design a DFA over $\{0, 1\}$ that accepts all strings with an even number of 0s.
- 7) Construct a DFA over $\{a, b\}$ that accepts all strings that start with a and end with b .
- 8) Write a regular expression for all strings over $\{a, b\}$ that do not contain the substring bb .
- 9) Prove that the language $L=\{w \in \{0,1\}^* \mid w \text{ contains exactly one } 1\}$ is regular
- 10) Prove that the language $L=\{0^n 1^n \mid n \geq 0\}$ is not regular.
- 11) Construct a regular expression for all strings over $\{a, b\}$ that contain both a and b .
- 12) Given the language $L=(ab)^*$ draw a DFA and write regular expressions, and justify whether L is regular or not.